Hidden Groundwater Contamination Patterns in Tamil Nadu Using Python

An Exploratory and Statistical Investigation of Groundwater Chemical Quality Across Tamil Nadu

1. Project Overview

This project focuses on analysing groundwater quality measurements collected by the Central Ground Water Board (CGWB) across Tamil Nadu.

Using Python, we will examine chemical parameters such as pH, fluoride, nitrate, arsenic, uranium, total dissolved solids (TDS), electrical conductivity (EC), and hardness.

The project will investigate which districts have elevated chemical measurements, which parameters are related, whether multiple chemicals exceed reference limits in the same sample, and how groundwater quality varies over time.

The final goal is to transform raw government monitoring records into meaningful statistical findings, visualizations, and evidence-based monitoring recommendations.

2. Data Source

Official Indian Government Dataset

Ground Water Quality — Manual Chemical Parameters (CGWB), Tamil Nadu

Department	Central Ground Water Board, Ministry of Jal Shakti
Portal	National Water Data Portal
Geographic coverage	Tamil Nadu, India
Listed data period	1961–2025
File format	CSV
File size	Approximately 1.8 MiB, as previously identified
Records and attributes	Exact counts pending CSV inspection

Open / Download Tamil Nadu CSV

Dataset validation: Once downloaded, we'll use df.shape, df.columns, and df.info() to confirm at least 500 records and 10 attributes, as required by the rubric. The published date range does not guarantee observations exist in every year.

3. Problem Statement

Groundwater chemical quality can differ across regions because of geological conditions, land use, human activities, and other environmental factors.

Government monitoring data contains numerous chemical measurements, but these observations are difficult to interpret without cleaning, statistical analysis, and comparison across locations and time.

This project aims to answer:

Which districts and monitoring stations in Tamil Nadu show elevated groundwater chemical concentrations, how are different chemical parameters associated, and which areas may benefit from additional monitoring?

The analysis will identify patterns and possible contamination concerns, but it will not claim to establish the causes of contamination without supporting evidence.

4. Project Objectives

1. Understand the groundwater dataset's structure, coverage, and data quality.
2. Clean missing, duplicate, inconsistent, and invalid observations.
3. Calculate descriptive statistics for major groundwater chemical parameters.
4. Identify the distribution and variability of fluoride, nitrate, arsenic, uranium, pH, TDS, and hardness.
5. Compare chemical measurements across districts and monitoring stations.
6. Analyse correlations between groundwater chemical parameters.
7. Identify samples exceeding selected applicable water-quality reference thresholds.
8. Investigate whether multiple elevated chemical parameters occur together.
9. Examine annual or seasonal changes where the available data supports comparisons.
10. Develop at least 15 meaningful visualizations and derive five or more evidence-based findings.

5. Attribute (Column / Feature) Details

These are the analysis fields we intend to use from the official resource. We'll map them to the exact CSV headers after downloading it.

Attribute	Description	Analytical purpose

State	State name	Tamil Nadu filtering
District	Administrative district	District comparison
Station / Location	Monitoring point	Station-level trends
Latitude	Geographic coordinate	Mapping
Longitude	Geographic coordinate	Mapping
Sampling Date	Collection date	Temporal analysis
pH	Acidity/alkalinity measure	pH distribution
EC	Electrical conductivity	Mineralization indicator
TDS	Total dissolved solids	Dissolved-solids analysis
Fluoride	Fluoride concentration	Elevated fluoride screening
Nitrate	Nitrate concentration	Elevated nitrate screening
Arsenic	Arsenic concentration	Trace-element analysis
Uranium	Uranium concentration	Trace-element analysis
Total Hardness	Hardness measurement	Mineral content
Calcium	Calcium concentration	Mineral composition
Magnesium	Magnesium concentration	Mineral composition
Chloride	Chloride concentration	Chemical composition
Sulphate	Sulphate concentration	Chemical composition

Some measurements may have missing values, different units, or incomplete temporal coverage. Only verified, usable columns will be included in the final calculations.

6. Tools & Technologies

Technology	**Usage**
Python	Complete analytical workflow

Pandas	Data loading, cleaning, grouping, pivot tables
NumPy	Numerical calculations and derived features
Matplotlib	Histograms, bar charts, line charts
Seaborn	Box plots, heatmaps, statistical comparisons
Plotly	Interactive geographic and exploratory charts
Jupyter Notebook / Colab	Executable notebook and documentation

7. Data Preprocessing and Feature Engineering

A. Data cleaning steps

We will:

- Load the CSV using pd.read_csv().
- Inspect rows, columns, data types, and descriptive statistics.
- Identify missing values for every field.
- Check duplicate records using appropriate station, date, and measurement identifiers.
- Standardize district and station names.
- Convert sampling dates into datetime.
- Convert chemical columns into numeric data.
- Check invalid pH values, negative concentrations, and unit inconsistencies.
- Investigate extreme values before deciding whether they are valid observations or errors.
- Preserve missing measurements as missing rather than automatically replacing them with zero.

B. Feature engineering

Derived column	**How we'll calculate it**
sampling_year	Extract year from sampling date
sampling_month	Extract month from sampling date

sampling_season	Group sampling months into defined seasonal categories
ph_category	Acidic, neutral, or alkaline classification
fluoride_exceedance	Whether fluoride exceeds the selected reference threshold
nitrate_exceedance	Whether nitrate exceeds the selected reference threshold
arsenic_exceedance	Whether arsenic exceeds the selected reference threshold
uranium_exceedance	Whether uranium exceeds the selected reference threshold
exceedance_count	Number of assessed parameters exceeding their reference limits
multi_parameter_flag	Whether two or more assessed parameters exceed thresholds

Thresholds will be taken from relevant Indian drinking-water standards, with parameter units and applicable limits documented. We won't treat unmeasured parameters as safe or classify a groundwater source as definitively drinkable based only on selected chemicals.

8. Analysis and Visualizations

Our EDA will be organized into three categories:

5

Univariate charts

5

Bivariate charts

5

Multivariate charts

We will also prepare summary tables and statistical interpretations to explain each important chart.

9. Exploratory Data Analysis (EDA)

Before visualization, we will calculate:

- Number of samples, districts, and unique monitoring stations.
- Earliest and latest valid sampling dates.
- Number of observations per year and district.

- Percentage of missing values in each chemical parameter.
- Distribution of chemical concentrations.
- Number of repeated observations at each station.
- District-wise median chemical concentrations.
- Proportion of assessed samples above selected thresholds.

This stage will determine which variables are sufficiently complete for meaningful analysis.

10. Statistical Analysis

We will calculate:

Statistic	**Application**
Mean	Average concentration
Median	Typical concentration less affected by extremes
Mode	Most frequent category, where meaningful
Minimum	Lowest observed value
Maximum	Highest observed value
Standard deviation	Variability
Variance	Dispersion
Quartiles / IQR	Spread and potential outliers
Pearson correlation	Linear relationships
Spearman correlation	Rank-based relationships
Groupby summaries	District- and year-wise comparisons
Pivot tables	District × parameter and year × district comparisons

For highly skewed chemical concentrations, medians and percentiles may be more informative than averages.

11. Data Visualization

The following is our exact planned chart list, with the research question and interpretation we'll produce after analysing the CSV.

A. Univariate Analysis — 5 charts

1. Distribution of Groundwater pH — Histogram

What we'll do: Plot pH frequencies and identify the typical range.

Interpretation we'll write: Whether most measured samples are acidic, neutral, or alkaline, and whether extreme pH observations occur.

2. Distribution of Fluoride — Histogram

What we'll do: Plot fluoride concentration, with a reference threshold if units and standards are confirmed.

Interpretation: Whether concentrations are concentrated at low values or show a long tail of elevated measurements.

3. Distribution of Nitrate — Box Plot

What we'll do: Examine the median, quartiles, spread, and extreme nitrate values.

Interpretation: Whether nitrate concentrations vary substantially and which observations warrant investigation.

4. Distribution of Arsenic — Histogram

What we'll do: Plot arsenic values after checking missing data and detection-limit notation.

Interpretation: Whether measurements are mostly low or include elevated observations.

5. Monitoring Sample Count by District — Bar Chart

What we'll do: Count valid samples across Tamil Nadu districts.

Interpretation: Which districts are represented more heavily and where monitoring coverage is comparatively limited.

B. Bivariate Analysis — 5 charts

6. District vs Fluoride Concentration — Box Plot

What we'll do: Compare fluoride concentration distributions across districts.

Interpretation: Identify districts with higher typical values, greater variability, or unusually high observations.

7. District vs Nitrate Concentration — Bar Chart

What we'll do: Calculate district-wise median nitrate concentrations.

Interpretation: Compare typical nitrate levels, while considering sample counts.

8. Electrical Conductivity vs TDS — Scatter Plot

What we'll do: Plot EC against TDS and calculate their correlation.

Interpretation: Assess the strength and direction of their association, while checking whether TDS was directly measured or estimated from EC.

9. Sampling Year vs Median Fluoride — Line Chart

What we'll do: Calculate annual median fluoride values, alongside yearly sample counts.

Interpretation: Investigate temporal variation without assuming that changes in station coverage represent true environmental trends.

10. District vs Fluoride Exceedance Rate — Bar Chart

What we'll do: Calculate the percentage of valid fluoride samples exceeding the selected threshold in each district.

Interpretation: Identify districts with comparatively high observed exceedance rates.

C. Multivariate Analysis — 5 charts

11. Chemical Parameter Correlation — Heatmap

What we'll do: Calculate correlations among pH, EC, TDS, fluoride, nitrate, hardness, calcium, magnesium, and other sufficiently complete measurements.

Interpretation: Identify strong positive or negative relationships, without treating correlation as causation.

12. District × Chemical Parameter — Heatmap

What we'll do: Create a matrix of district-wise median concentrations for selected parameters, standardized for visualization.

Interpretation: Compare chemical profiles across districts and identify distinctive combinations of elevated measurements.

13. Fluoride vs Nitrate by District — Scatter Plot

What we'll do: Plot paired fluoride and nitrate observations, with district grouping.

Interpretation: Determine whether elevated fluoride and nitrate tend to appear together in the same samples.

14. Multi-Parameter Exceedance by District — Stacked Bar Chart

What we'll do: Group samples into zero, one, or multiple assessed threshold exceedances, using comparable complete-case measurements.

Interpretation: Compare how frequently multiple elevated chemical indicators occur together across districts.

15. Groundwater Chemical Quality — Interactive Geographic Map

What we'll do: Plot valid station coordinates in Plotly, with a selectable or predefined chemical measurement represented by marker colour.

Interpretation: Explore the geographic distribution of elevated observations and potential monitoring priorities.

The images illustrate visualization styles only. Actual charts, numbers, and interpretations will be generated from the Tamil Nadu CSV.

12. Summary of Findings

The completed project will contain at least five findings, supported by calculations and charts.

Our primary findings will address:

1. Geographic variation: Which districts have higher median fluoride and nitrate measurements?
2. Threshold exceedances: Where are elevated measurements most frequently observed among tested samples?
3. Chemical relationships: Which parameters show the strongest correlations?
4. Multi-parameter patterns: How often do selected chemical exceedances occur together?
5. Monitoring coverage: Which districts or stations have relatively limited data?
6. Temporal variation: Do comparable monitoring observations show changes across years?

7. Anomalies: Are there unusual concentrations that merit closer investigation?

We will report actual district names, statistics, percentages, and chart references after running the analysis. No findings will be invented in advance.

13. Types of Analysis

Type	Exactly what we'll do
Descriptive	Summarize samples, chemical distributions, district medians, and exceedance percentages
Diagnostic	Investigate whether elevated values are associated with districts, sampling periods, or other chemical parameters
Predictive	Optional extension: estimate a selected chemical parameter using other measurements, if sufficient reliable data exists
Prescriptive	Recommend monitoring priorities, repeat sampling, and further laboratory investigation based on observed patterns

We will not claim to have identified the causes of contamination merely from statistical associations.

14. Future Enhancements and Suggestions

Potential enhancements after the final project include:

- Expanding the analysis to neighbouring states.
- Incorporating land-use, geological, or groundwater-depth data.
- Developing a Power BI or Plotly Dash interactive dashboard.
- Applying suitable machine-learning models to larger validated datasets.
- Automating data-quality checks and periodic monitoring reports.
- Comparing monitoring results across consistent sampling periods and stations.

15. Conclusion

This project will demonstrate a complete Python data analytics workflow using an official Indian government groundwater dataset.

Through data cleaning, statistical analysis, and 15 planned visualizations, the study will investigate chemical-quality variation, elevated measurements, correlations, and geographic monitoring patterns across Tamil Nadu.

The final conclusion will summarize actual findings and explain how the analysis can support evidence-based groundwater monitoring and further investigation.